Reproduction and Critical Evaluation of Representation-State Adaptive RAG

Field	Recorded value
Evaluation date	3 August 2026
Status	Completed implementation audit and paired feasibility pilot; confirmatory evaluation not completed
Models actually used	`sentence-transformers/all-MiniLM-L6-v2` for embeddings and Groq `openai/gpt-oss-120b` for language-model operations

Executive conclusion

The current experimental evidence does **not** support the claim that the proposed method outperforms Traditional RAG overall.

The code executes end to end and its central mechanism worked on the one answerable QASPER pilot: the proposed method selected hierarchical and semantic representations, recovered the annotated evidence, and improved answer token F1 from 0.027 to 0.240. The fixed-k baseline did not recover the annotated evidence. On one unanswerable MuSiQue pilot, both systems correctly abstained.

These are useful feasibility observations, not research results. Only two examples were completed, answer quality and retrieval quality each have only one scoreable example, the baseline k=5 was not calibrated, the methods used unequal context, and faithfulness and answer relevancy were not independently measured. The proposed method also averaged 45.5 Groq calls, 473.8 seconds, 67,432 tokens, and $0.02153 per query, versus 1 call, 3.36 seconds, 1,679 tokens, and $0.000446 for Traditional RAG.

The correct research decision is therefore: **promising retrieval behavior, serious efficiency problems, and insufficient evidence for superiority**.

1. What was actually reproduced

The following checks completed successfully:

- A project-local `.venv` was created from the lock file.
- `all-MiniLM-L6-v2` loaded locally through Transformers/PyTorch and returned a normalized 384-dimensional embedding.
- Eleven deterministic unit tests passed.
- The Traditional RAG baseline executed with one sliding-window dense index and static k=5.
- The proposed five-view adaptive pipeline executed with sentence-window, sliding-window, paragraph, semantic, and hierarchical indexes.
- One paired QASPER validation example and one paired MuSiQue-Full validation example completed.

- API latency, wall latency, calls, prompt/completion/cached tokens, and estimated Groq cost were logged.
- Checkpoints, JSON results, CSV metrics, and comparison graphs were produced.

The following requested work did **not** complete and must not be described as completed:

- Full held-out evaluation over QASPER test and MuSiQue-Full development.
- Training-only hyperparameter calibration.
- Custom representation-state benchmark evaluation.
- HotpotQA external-generalization evaluation.
- Independent human or external-model faithfulness and answer-relevancy scoring.
- Repeated runs to measure model/API variability.

Two infrastructure failures were excluded from metrics: a missing runtime package during a first invocation and a missing plotting dependency after MuSiQue inference. The latter did not lose inference results. Failed or partial calls were not counted as successful examples.

2. Systems compared

Traditional RAG baseline

1. Split the known source document into fixed 256-word sliding windows with 64-word overlap.
2. Embed each window with `all-MiniLM-L6-v2`.
3. Embed the query with the same model.
4. Retrieve a fixed top `k=5` by dense similarity.
5. Keep evidence under a 2,000-word context limit.
6. Ask the same Groq answer model to answer or declare insufficient evidence.

This baseline has no representation probe, RRF selection, LLM evidence judge, adaptive continuation, view reactivation, reformulation, or post-retrieval cross-view deduplication.

Proposed Representation-State Adaptive RAG

1. Parse the source into canonical units with stable provenance.
2. Build five representation indexes: sentence-window, sliding-window, paragraph, semantic, and hierarchical.
3. Retrieve only the top three from each representation as a **probe**. This depth is for representation selection, not final retrieval.
4. Extract the query's evidence requirements.

5. Judge probe candidates against those requirements.
6. Align overlapping candidates and calculate rank-only RRF support. RRF helps choose representations; it is not an answer score.
7. Select representations using evidence obligations, complementary coverage, RRF support, and evidence cost.
8. Continue retrieving from selected indexes in batches without a fixed final k.
9. Stop only after coverage and an empty confirmation batch, or after bounded reactivation/reformulation proves insufficiency.
10. Remove exact and conservative semantic duplicates **after** adaptive retrieval.
11. Compact evidence within the context budget.
12. Generate a cited answer or a deterministic insufficiency outcome.

The final k is the number of evidence items remaining after stopping, deduplication, and compaction. It is an output, not a tuned retrieval target.

3. Data used in this pilot

Dataset	Split	Completed items	Role
QASPER v0.3	Validation/development	1 answerable	Long scientific document and evidence retrieval smoke test
MuSiQue-Full v1.0	Validation/development	1 unanswerable	Insufficiency and abstention smoke test

The files were deterministic first-record samples. This is convenient for debugging but not representative sampling. No conclusion may be generalized to either dataset.

The intended confirmatory locally scoreable population is QASPER test plus MuSiQue-Full development. MuSiQue test labels are hidden and cannot be used for local scoring.

4. Reproduced results

4.1 Answer and evidence quality

Metric	Traditional RAG	Proposed RAG	Paired n	Interpretation
Answer exact match	0.000	0.000	1	Neither generated answer exactly matched the short QASPER references
Answer token F1	0.027	0.240	1	Proposed answer overlapped the QASPER annotations more strongly
Evidence-set precision	0.000	0.100	1	Proposed included one matching evidence item among ten
Evidence-set recall	0.000	1.000	1	Proposed recovered the annotated QASPER evidence; baseline did not
Retrieval MRR	0.000	0.143	1	First matching proposed evidence appeared at rank 7
Answerability accuracy	1.000	1.000	2	Both classified one answerable and one insufficient example correctly
Answerability precision	1.000	1.000	2	Descriptive only
Answerability recall	1.000	1.000	2	Descriptive only
Answerability F1	1.000	1.000	2	Descriptive only

Retrieval metrics are undefined on the MuSiQue example because it has no gold relevant evidence. The original evaluator incorrectly treated no-gold recall and nDCG as 1.0; the corrected evaluator records them as `NaN` and excludes them from retrieval means.

4.2 QASPER retrieval at cutoffs

Metric	Traditional RAG	Proposed RAG
Precision@5	0.000	0.000
Recall@5	0.000	0.000
nDCG@5	0.000	0.000
Precision@10	0.000	0.100
Recall@10	0.000	1.000
nDCG@10	0.000	0.333
Precision@20	0.000	0.050
Recall@20	0.000	1.000
nDCG@20	0.000	0.333

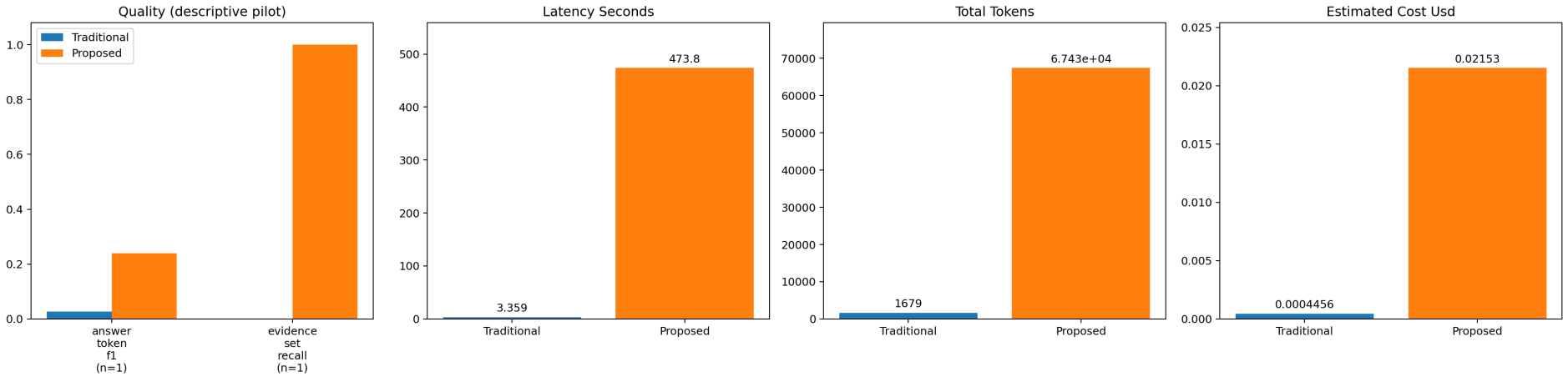
These values measure the ordered **final evidence set**, not the complete raw candidate ranking. Metrics at cutoffs larger than the final evidence count therefore require careful interpretation.

4.3 Efficiency

Metric, mean over 2 queries	Traditional RAG	Proposed RAG	Proposed / Traditional
Query wall latency	3.36 s	473.77 s	141.0x
Groq calls	1.0	45.5	45.5x
Prompt tokens	1,248.5	40,621.5	32.5x
Completion tokens	430.5	26,810.5	62.3x
Total tokens	1,679	67,432	40.2x
Estimated cost/query	$0.000446	$0.021527	48.3x
Final evidence count	5.0	5.0	Not meaningful as an average
Final evidence words	596.5	923.0	1.55x

Cost uses the recorded API token counts and the current published GPT-OSS 120B rates: $0.15 per million uncached input tokens, $0.075 per million cached input tokens, and $0.60 per million output tokens. It excludes failed setup attempts, network overhead, local compute, and the skipped external quality judge.

Latency is query-time wall latency and includes rate-limit waits. Index construction latency was not recorded in these pilots, so the table is not complete end-to-end document-ingestion latency.



Pilot comparison

5. Statistical significance

No performance improvement is statistically established.

Metric	Proposed minus Traditional	Paired n	Randomization p-value	Decision
Answer token F1	+0.213	1	1.000	Not testable meaningfully
Evidence-set recall	+1.000	1	1.000	Not testable meaningfully
Answerability correctness	0.000	2	1.000	No observed difference
Latency	+470.41 s	2	0.497	Not significant; large descriptive penalty
Total tokens	+65,753	2	0.495	Not significant with n=2; consistently much larger
Estimated cost	+$0.02108	2	0.493	Not significant with n=2; consistently much larger

The bootstrap intervals for answer quality are degenerate because there is one scoreable item. With two paired observations, a sign-randomization test has extremely coarse resolution. Reporting `p > 0.05` here does not mean the systems are equivalent; it means the experiment is underpowered.

The confirmatory study should use paired bootstrap confidence intervals for EM/F1/retrieval metrics, McNemar's exact test for paired answerability decisions, and paired randomization or Wilcoxon tests for latency/token/cost distributions. Correct multiple primary comparisons using Holm's method and report effect sizes, not only p-values.

6. Failure-case analysis

QASPER answerable example

The question asked which multilingual approaches were compared. Traditional RAG retrieved nearby material about multilingual methods, but not the annotated comparison evidence. Its answer listed several plausible but off-target methods. Proposed RAG selected hierarchical and semantic views, continued to ten final items, recovered the gold span at rank 7, and answered with pivoting, multilingual NMT, and cross-lingual transfer.

Why the proposed method helped: the relevant statement appears to need both topical similarity and paper structure. A static five-window cutoff stopped before it. Adaptive continuation plus hierarchical/semantic selection reached it.

Why the proposed score was still low: QASPER annotations for this item are short reference labels such as `BIBREF19` and `BIBREF20`, while the generated answer used method names and citations. Exact match therefore penalizes a semantically useful answer. The official QASPER answer evaluator and annotation-aware reference normalization should be added before publication.

MuSiQue insufficient-evidence example

Both systems correctly declared the question unanswerable from the supplied context. Traditional RAG did so in one call. The proposed method selected/reactivated all five views, made 50 calls, exhausted its evidence search, and returned an empty answer with `INSUFFICIENT` and final `k=0`.

Why the proposed method was worse operationally: proving absence caused repeated judging, view reactivation, and reformulation. This is a predictable worst case for the current design. The empty answer is also less helpful to a user than the baseline's short explanation of the missing fact, even though both answerability labels are correct.

7. Implementation and evaluation issues found

Severity	Issue	Consequence	Required correction
Critical	Only two smoke examples completed	No generalization or superiority claim	Run the frozen full confirmatory splits
Critical	Baseline k=5 was not selected on training data	Potentially unfair quality/context comparison	Calibrate static k on train only and also report a context-matched baseline
Critical	Proposed method used roughly 1.55x final context words and many more calls	Quality gains may come from more compute/context	Report both conventional and resource-matched comparisons
High	Same Groq model family performs proposed routing/judging and answer generation	Correlated self-confirmation risk	Use blinded humans or a separately validated external judge for secondary quality scoring
High	Faithfulness and answer relevancy were skipped	Requested metrics are missing	Score stored evidence/answers with blinded independent evaluation
High	Retrieval relevance uses normalized text containment	Paraphrases may be missed; partial spans can be misclassified	Add dataset-native evidence IDs and official support evaluators
High	Retrieval metrics are computed on final evidence, not full rankings	P@k, Recall@k, MRR, nDCG are not standard candidate-ranking measures	Preserve raw ranked candidates before judging/deduplication and score both stages separately
High	Index-build latency was not logged	End-to-end latency is incomplete	Measure parsing, chunking, embedding, indexing, query retrieval, judging, and generation separately
High	Actual run uses five views while one research specification describes three primary views	Experimental condition is ambiguous	Make the five-view condition authoritative or preregister three-view primary plus five-view ablation
High	Dataset decision documents disagree about whether ConditionalQA is core	Risk of selective benchmark reporting	Freeze one protocol before confirmatory execution
Medium	MuSiQue no-gold retrieval metrics were initially counted as perfect	Inflated retrieval means	Fixed: record undefined metrics as `NaN`
Medium	Baseline citations were requested but not validated	Invalid evidence indices could pass	Fixed: validate citation presence/range for answerable outputs
Medium	First-N smoke samples are not stratified or random	Selection bias	Use a recorded seed and stratification only on train for calibration
Medium	No repeated API runs	Unknown nondeterminism/reliability	Repeat a prespecified subset three times and report variance
Medium	API call checkpointing occurs per example, not per call/stage	A late failure can waste many calls	Cache structured stage outputs by prompt/model hash
Medium	Context limit is counted in words, not model tokens	Budget differs from actual API tokens	Use the model tokenizer for hard context budgets; keep words as a secondary readable measure

No direct insertion of gold answers/evidence into prompts was observed in the execution path. However, automatic checks against MuSiQue's published source-question leakage registry are not enforced by the adapter, and pretrained-model benchmark exposure cannot be ruled out. This must be disclosed.

8. Calibration and final evaluation protocol

No final retrieval `k` is calibrated for the proposed method. Its `final_k` remains emergent. The following decisions should be frozen **before** examining validation/test outcomes.

Training-only calibration data

- QASPER train for long-document/hierarchy controls.
- MuSiQue-Full train, stratified by answerability and hop count, for sufficiency controls.
- A recorded random seed (`20260803`).
- Stratify additionally by document length and gold evidence count where available.

Traditional baseline calibration

Evaluate `k in {3, 5, 8, 10, 15, 20}` on training only, under the same 2,000-token final context ceiling. Select the smallest `k` whose macro-average evidence recall is within one percentage point of the best grid value. This avoids choosing an unnecessarily expensive `k` while preventing validation/test tuning.

Report two baseline variants:

1. **Traditional calibrated baseline:** training-selected static `k`.
2. **Resource-matched baseline:** same final context-token ceiling and a Groq-call budget matched as closely as practical to the proposed system. This separates representation/state logic from simply using more computation.

Proposed-method calibration

Keep the primary controls already specified by the implementation: probe depth 3, RRF constant 60, adaptive batch size 3, maximum context budget, bounded candidate count, bounded retrieval rounds, empty-confirmation stopping, view reactivation, reformulation, and post-retrieval deduplication. Tune thresholds or budgets only on train. Probe depth 5 and alternative view sets belong in ablations, not post-hoc replacements.

Confirmatory datasets

- QASPER test: final public long-document result.
- MuSiQue-Full development: final locally scored answerability/multi-hop result because test labels are hidden.
- Custom representation-state benchmark: required before claiming correct state diagnosis or representation selection.
- HotpotQA: optional external-generalization result, clearly separate from the primary claim.

Primary outcomes

Predeclare a small primary family: QASPER answer F1, QASPER evidence recall/F1, MuSiQue answerability F1, MuSiQue support/hop recall, and total query cost. Treat other measures as secondary. This prevents cherry-picking from many metrics.

9. Does the evidence support superiority?

No. The evidence supports only these narrow statements:

1. The implementation can execute the intended no-fixed-final-k pipeline.
2. On one QASPER example, adaptive representation selection recovered evidence missed by static top-5 retrieval.
3. On one MuSiQue unanswerable example, both systems abstained correctly.
4. In both completed examples, the proposed method consumed dramatically more time, calls, tokens, and estimated cost.

It does not establish higher average accuracy, precision, recall, F1, faithfulness, relevancy, or retrieval quality on a population. It also does not establish that RRF-based representation selection is the cause of the observed QASPER gain; ablations are required.

10. Final research recommendation

The method has a coherent research idea: use a small cross-representation probe to diagnose which representations contain complementary evidence, then retrieve adaptively until evidence requirements are satisfied, and deduplicate only afterward. This directly addresses the weakness of choosing one chunk representation and one fixed final k for every question.

The current implementation is too call-intensive for a full-scale confirmatory run under ordinary Groq rate limits. Before spending on the full study:

1. Batch evidence judgments or judge a compact candidate table in one call per round.
2. Use deterministic embedding/rule signals to reject obviously irrelevant probe candidates before LLM judging.
3. Cache requirement extraction, candidate judgments, and reformulations by content hash.
4. Add an early, calibrated insufficiency gate to prevent 50-call absence proofs.
5. Preserve raw rankings and official dataset evidence identifiers.
6. Calibrate and freeze the baseline and proposed budgets on train only.
7. Run ablations: no RRF, no hierarchical view, single best view, no reactivation, no reformulation, and fixed-k over the selected views.
8. Only then execute the held-out paired study and independent quality evaluation.

Until those steps are completed, the defensible paper conclusion is: **the approach is technically feasible and produced one encouraging evidence-retrieval case, but there is not yet sufficient, fair, or statistically meaningful evidence that it outperforms Traditional RAG.**

Reproduction artifacts

- `results.json`: corrected combined pilot metrics and raw records.
- `paired_metrics.csv`: per-example/per-method metrics.
- `comparison.png`: quality and efficiency graph.
- Individual completed runs remain in `output/experiments/pilot_qasper_1_retry` and `output/experiments/pilot_musique_1`.
- Groq pricing reference: https://console.groq.com/docs/model/openai/gpt-oss-120b